Case Study — KABBALAH Helps Discover Design Questions, Analyze Optimization Processes, and Formulate and Test Hypotheses

> **Purpose of this case study.** This example is not intended to prove a universal P–C design law. Its purpose is to show how a historical design trajectory can expose a design question, support progressively deeper hypotheses, and finally convert qualitative designer experience into a quantitative and falsifiable description.

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1. Route divergence → a design question

Both routes originate from the same reconstructed spherical structure, **ST000032**, and both eventually reach the target **F/1.7**. The key structural intervention that distinguishes the two routes is whether the glass materials of **L3 and L4** are exchanged — equivalently, whether the glass assignment at **S7 and S9** is swapped.

- **No swap:** the original L3/L4 glass ordering is retained.
- **Swap:** after opening the Qcon degrees of freedom and performing AUT, the glass materials assigned to **L3 (S7)** and **L4 (S9)** are exchanged, followed by further optimization.

This L3/L4 glass-material exchange is therefore the central route-level difference examined in this case study. The two routes then evolve toward markedly different final structures and nominal performance.

Route identification in the KABBALAH trajectory

- **No swap:** `ST000032 → ST000221 → … → ST000314`
- **Swap:** `ST000032 → ST000350 → … → ST000497/498`

> **Figure note for the final presentation:** highlight these two branches directly in the KABBALAH path map, and mark the **L3/L4 (S7/S9) glass swap** as the key intervention separating the routes.

Common starting structure — ST000032

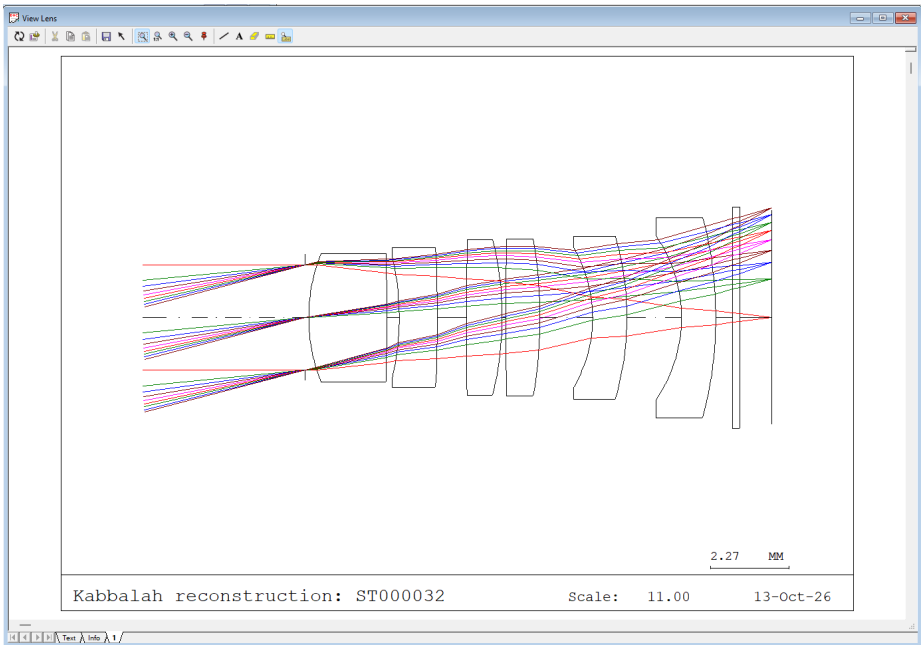
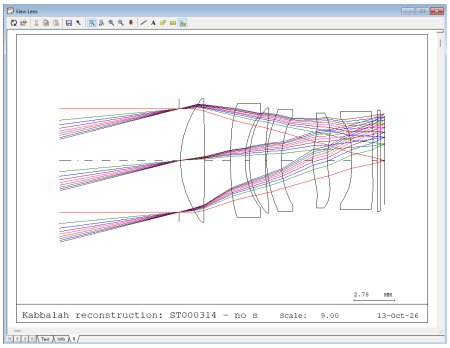
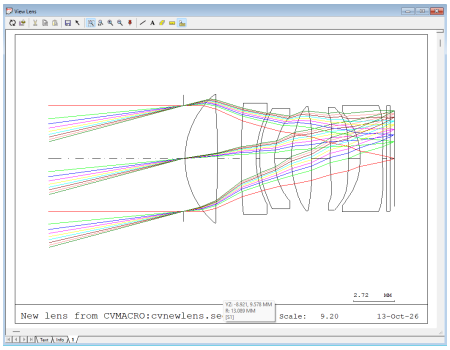


Figure 1: Common starting structure: ST000032

No swap — final ST000314	Swap — final ST000497/498
	

Final nominal-performance comparison

Metric	No swap	Swap
F/#	1.7	1.7
EFL	12 mm	12 mm
TTL	13.5 mm	14.0 mm
Field	3.21 mm real image height	3.21 mm real image height

Metric	No swap	Swap
Max. distortion	$< 6\%$	$\mathbf{< 0.2\%}$
MTF @ 300 lp/mm	0.25	**0.48**
Min. Strehl ratio	0.48	**0.74**
Relative illumination	> 0.60	$\mathbf{> 0.89}$

During the **No-swap** trajectory, repeated recovery of marginal-ray height required comparatively large structural corrections and was accompanied by a substantial field-curvature cost. This motivates the first question:

> **Question 1 — Does the No-swap route develop a systematically larger ray-height burden as the aperture is opened?**

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2. Hypothesis 1 — the No-swap route develops a larger downstream ray-height burden

To test the first question, the marginal-ray projection height at the **front surface of L3 (S7)** was compared along the two routes.

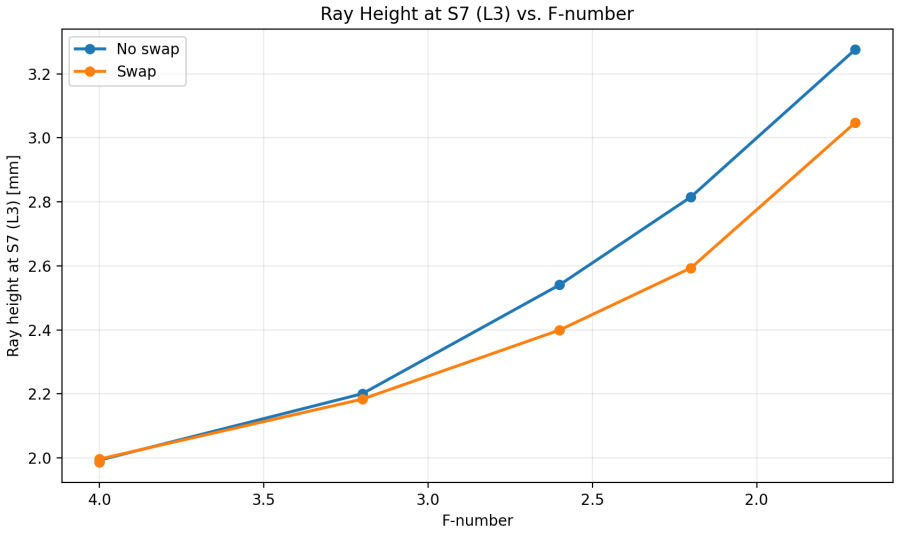


Figure 2: Ray height at S7 versus F-number

F/#	No swap: S7 ray height [mm]	Swap: S7 ray height [mm]
3.2	2.2007	2.1839

F/#	No swap: S7 ray height [mm]	Swap: S7 ray height [mm]
2.6	2.5406	2.3994
2.2	2.8154	2.5937
1.7	3.2758	3.0476

The two routes are initially very similar, but the difference grows progressively as F/# decreases. The trajectory data therefore support the first hypothesis:

> **Hypothesis 1 is supported:** the No-swap route develops a larger downstream ray-height burden at L3.

This immediately raises a deeper question:

> **Question 2 — Why does the No-swap route generate the larger L3 ray height?**

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3. Hypothesis 2 — different allocation of Petzval and chromatic responsibilities

A possible explanation is that the middle group **L3–L4** does not carry the chromatic task in the same way in the two routes. If the L3–L4 group cannot economically establish local chromatic self-compensation, more compensation responsibility may be shifted to **L2**. A stronger negative L2 changes the outgoing ray angle and can increase the downstream projection height at L3:

$$\Delta y_3 \sim -d_{23} y_2 \Phi_2.$$

The following plots compare the evolution of **L2** and the **L3+L4 group** using Petzval contribution and system-weighted chromatic contribution.

No swap

Swap

The two trajectories show qualitatively different responsibility redistribution:

- **No swap:** L2 remains strongly involved in negative chromatic compensation, while the L3–L4 group shows no sustained development toward local self-compensation.
- **Swap:** as the aperture is opened, Petzval and chromatic responsibilities are progressively redistributed between L2 and L3–L4; the L3–L4 group develops substantially stronger internal compensation.

A classical chromatic-power argument also makes this interpretation physically plausible. In the original glass ordering, the approximate local color-neutrality condition

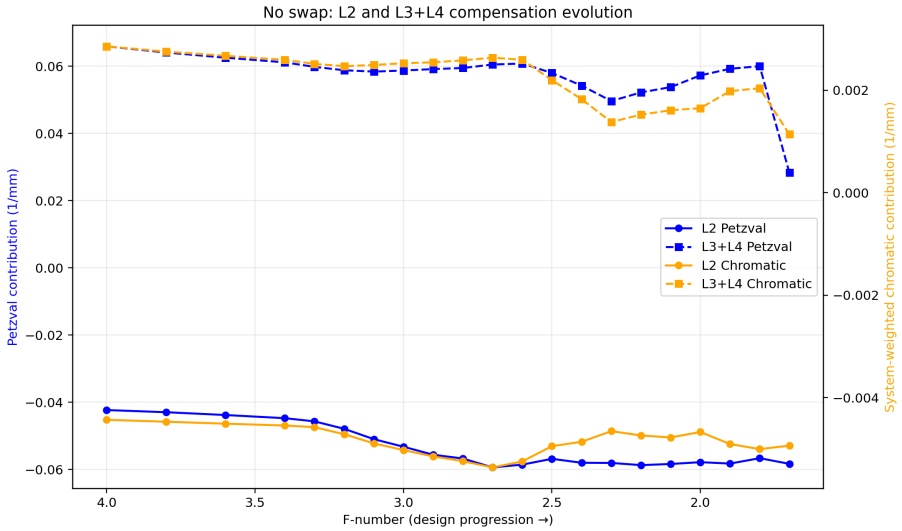


Figure 3: No swap P-C evolution

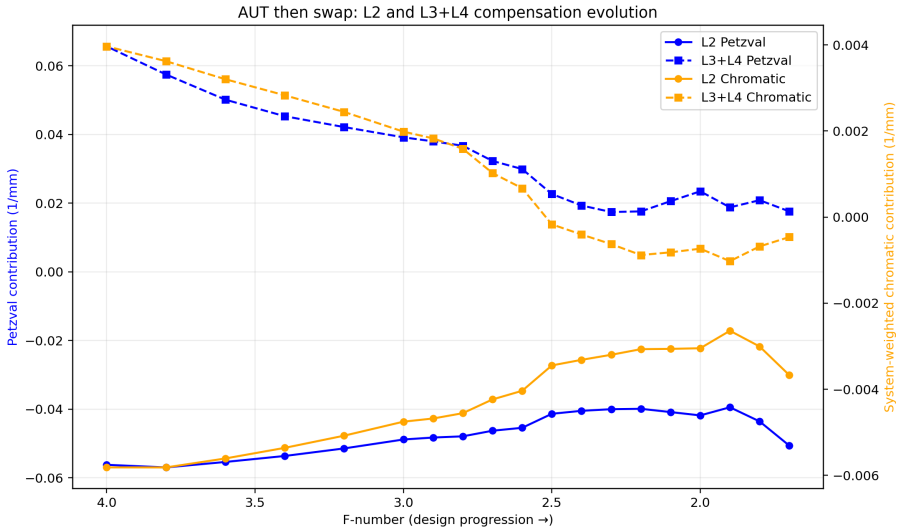


Figure 4: Swap P-C evolution

$$\frac{\Phi_3}{V_3} + \frac{\Phi_4}{V_4} \approx 0$$

with $V_3 \approx 55.8$ and $V_4 \approx 19.2$ implies approximately

$$|\Phi_3| \approx 2.9|\Phi_4|,$$

making an L3-negative / L4-positive local compensation expensive in Petzval terms. After the glass order is swapped, the corresponding negative-power requirement is much smaller (approximately 0.34 of the positive power), providing a more economical route to local chromatic compensation.

This leads to the case-study hypothesis:

> **Hypothesis 2** — Forming an **achromat-like local chromatic self-compensation structure** among neighboring optical elements **may increase the effective structural freedom of the subsystem under global chromatic constraints**, thereby enabling a more favorable redistribution of optical responsibilities during optimization.

A possible causal chain suggested by the trajectory is:

weak L3–L4 local compensation → larger compensation burden on L2 → larger ray angle after L2 → larger ra

States used in the P–C plots

No swap:
`ST000223, ST000224, ST000225, ST000226, ST000227, ST000229, ST000233, ST000240, ST000243, ST000245, ST000248, ST000252, ST000261, ST000267, ST000271, ST000274, ST000279, ST000292, ST000297, ST000301, ST000314`

Swap:
`ST000353, ST000357, ST000361, ST000366, ST000371, ST000375, ST000379, ST000384, ST000388, ST000391, ST000398, ST000403, ST000409, ST000426, ST000431, ST000435, ST000455, ST000458, ST000497`

> **State-selection rule:** use representative valid states along the active, non-rolled-back trajectory, approximately one final optimized state per aperture stage. State IDs are the authoritative references; in sections containing failed FNO attempts, the displayed F/# stage can differ slightly from the last physically valid aperture.

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4. From qualitative design experience to a quantitative trajectory language

The hypothesis above is still expressed in qualitative designer language — for example, *"the local group becomes more achromat-like," "compensation responsibility is redistributed,"* or *"the local functional role is being reorganized."* To make such observations comparable across routes and, eventually, usable by AI, a quantitative description is needed.

For a local optical group G, define the Petzval and system-weighted chromatic contributions as

$$P_G = \sum_{i \in G} P_i, \qquad C_G = \sum_{i \in G} C_{i,\mathrm{sys}},$$

where

$$C_{i,\mathrm{sys}} = \frac{\partial \Phi_{\mathrm{sys}}}{\partial n_i} \frac{n_i - 1}{V_i}.$$

4.1 Static internal-compensation descriptor: K

Define the normalized internal cancellation measures

$$\Gamma_P = \frac{|\sum_i P_i|}{\sum_i |P_i|}, \qquad \Gamma_C = \frac{|\sum_i C_i|}{\sum_i |C_i|},$$

and

$$\boxed{K = \sqrt{\Gamma_P \Gamma_C}}.$$

Interpretation:

- $K \to 1$: weak internal P–C cancellation;
- $K \to 0$: stronger internal P–C compensation.

K is a structural descriptor, not a quality score.

4.2 Dynamic evolution descriptor: θ and R_N

For successive states,

$$\Delta P_t = P_{t+1} - P_t, \qquad \Delta C_t = C_{t+1} - C_t,$$

and the normalized P–C evolution direction is

$$\theta_t = \operatorname{atan2}\left(\frac{\Delta C_t}{C_0}, \frac{\Delta P_t}{P_0}\right) \pmod{\pi}.$$

Directional coherence over a recent window is described by

$$\boxed{R_N = \left|\frac{1}{N}\sum e^{i2\theta_t}\right|}.$$

Thus:

$$\boxed{K = \text{current internal compensation state}}$$

$$\boxed{\theta,\ R_N = \text{recent evolution behavior}}$$

4.3 Provisional ABCD vocabulary

State	Quantitative meaning	Qualitative description
A	Low K, High R	strong internal compensation; coherent recent evolution
B	High K, High R	weak internal compensation; coherent recent evolution
C	High K, Low R	weak internal compensation; dispersed/changing evolution
D	Low K, Low R	strong internal compensation; local functional roles being redistributed

A/B/C/D are descriptive states, not quality labels.

The present case can be provisionally summarized as **No swap: C → B → C** and **Swap: B → A → D**, but these stage labels are currently qualitative interpretations rather than fixed classifications.

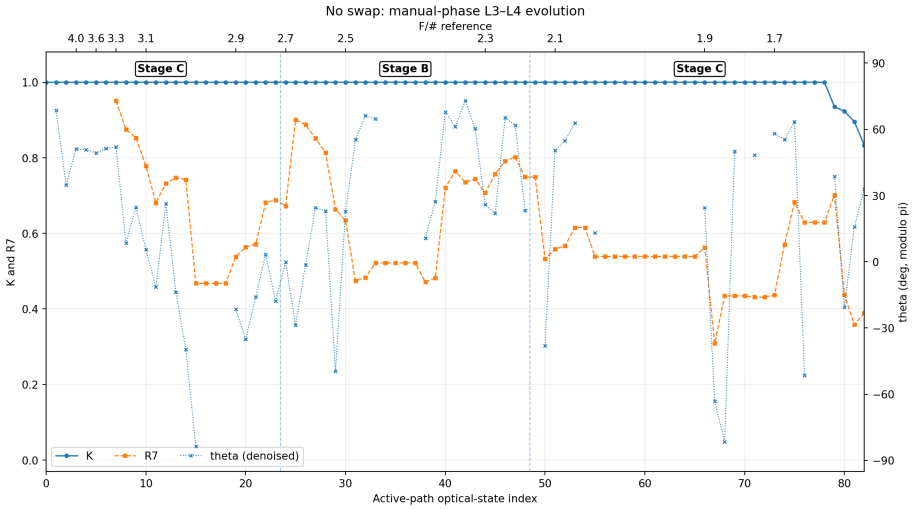


Figure 5: No swap K-R-theta

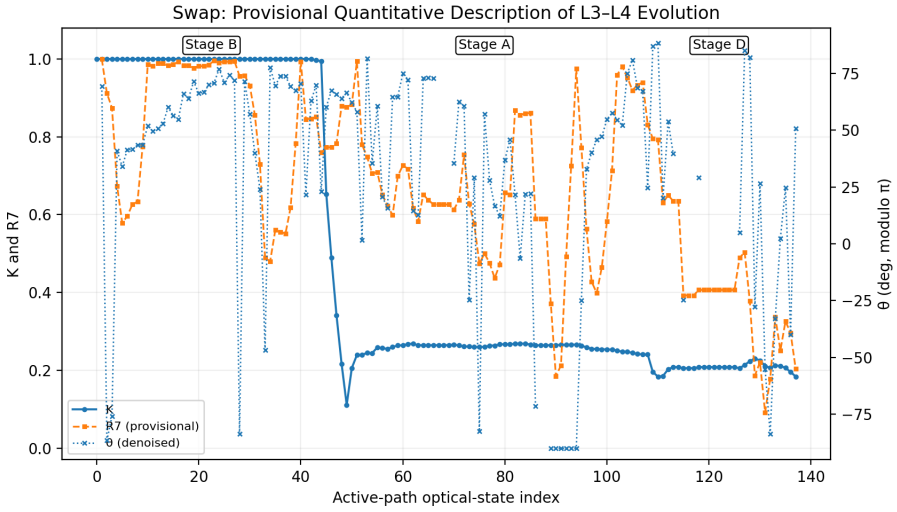


Figure 6: Swap K-R-theta

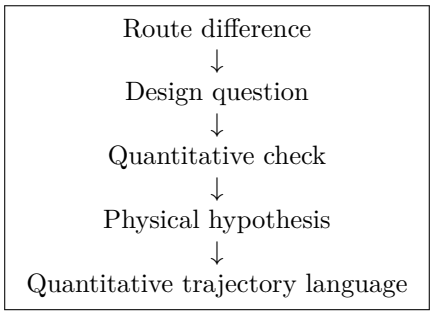
Provisional route visualizations

> **Important methodological limitation.** The current example uses a provisional recent-history window and manually interpreted ABCD phases. Before cross-design validation, the rule for the R_N window, negligible-transition filtering, and ABCD boundaries must be fixed independently of route outcome. A transition/uncertain zone around the classification boundaries should also be considered rather than forcing borderline points into A/B/C/D.

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5. What KABBALAH contributed in this case

The main result of this case study is not that a universal P–C law has already been proven. The methodological result is the sequence by which historical trajectory information was turned into a falsifiable design hypothesis:



KABBALAH did not directly generate the hypothesis. Instead, it preserved and reconstructed the historical differences needed to **discover the question, trace its origin, compare alternative routes, and formulate a quantitative description that can later be validated or rejected in independent designs**.

Next validation question

> **Across independent smartphone telephoto designs, do successful designs exhibit reproducible local P–C evolution patterns, or is the present ABCD behavior specific to this case?**

A negative result would still be informative: it would indicate that K, θ, and R_N are useful as a descriptive trajectory language but not necessarily as indicators of design quality. A positive result would motivate a stronger study of predictive trajectory signatures.